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Weather shield for a vehicle charge port

Abstract

A vehicle charge system includes a charge port having a charge port receptacle and a connector receiver. The charge port receptacle is configured to transfer electrical energy to a rechargeable electric storage system. A charge port shield is supported at the charge port. The charge port shield includes a base member having a peripheral edge defining a first surface and an opposing second surface. The base member includes an opening alignable with the charge port receptacle and a connector extending outwardly of the second surface. The connector being received by the connector receiver in the charge port to align the opening with the charge port receptacle. A frame extends about the peripheral edge of the base member. The frame includes a plurality of walls that project outwardly of the first surface of the base member. The plurality of walls and the first surface of the base member form a recess.

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Background/Summary

INTRODUCTION

(1) The subject disclosure relates to the art of electric vehicles and, more particularly, to a charge port weather shield for an electric vehicle.

(2) Electric vehicles include a charge port that is typically covered by a door. The door protects a charge receptacle from weather, dirt, and debris during driving and when parked outdoors. When charging, the door is opened and a charging device is connected to the charge receptacle. When the door is open, and the vehicle is charging out of doors, the charge receptacle may be exposed to weather. Rain, snow, and dirt accumulating between the charging device and the charge receptacle could affect charging efficiency. Accordingly, it is desirable to provide a device that shields vehicle charging components from the weather.

SUMMARY

(3) A vehicle charge system, in accordance with a non-limiting example, includes a charge port having a charge port receptacle and a connector receiver. The charge port receptacle is configured to transfer electrical energy from an electrical source to a rechargeable electric storage system (RESS). A charge port shield is supported at the charge port. The charge port shield includes a base member having a peripheral edge defining a first surface and an opposing second surface. The base member includes an opening alignable with the charge port receptacle and a connector extending outwardly of the second surface. The connector being received by the connector receiver in the charge port to align the opening with the charge port receptacle. A frame extends about the peripheral edge of the base member. The frame includes a plurality of walls that project outwardly of the first surface of the base member. The plurality of walls and the first surface of the base member form a recess.

(4) In addition to one or more of the features described herein the plurality of walls include a first wall, a second wall, a first side wall connecting the first wall and the second wall and a second side wall connecting the first wall and the second wall, the second wall, first side wall, and the second side wall extending from the first surface a first distance and the first wall extending from the first surface a second distance that is greater than the first distance.

(5) In addition to one or more of the features described herein the first wall defines an upper wall and the second wall defines a lower wall.

(6) In addition to one or more of the features described herein each of the first wall, the second

wall, the first side wall, and the second side wall extend from the first surface at an angle.

(7) In addition to one or more of the features described herein each of the first wall, the second wall, the first side wall, and the second side wall extend from the base member at an obtuse angle relative to the first surface.

(8) In addition to one or more of the features described herein each of the first wall, the second wall, the first side wall, the second side wall, and the first surface of the base member form an external charging device receiving cavity.

(9) In addition to one or more of the features described herein the connector receiver comprises a quick charge port.

(10) In addition to one or more of the features described herein the connector comprises a dummy quick charge connector.

(11) A vehicle, in accordance with a non-limiting example, includes a body defining a passenger compartment, an electric motor supported in the body, a rechargeable energy storage system (RESS) supported at the body. A charge port supported at the body. The charge port including a charge port receptacle connected to the RESS. A charge port shield is supported at the charge port. The charge port shield includes a base member having a peripheral edge defining a first surface and an opposing second surface. The base member includes an opening alignable with the charge port receptacle and a connector extending outwardly of the second surface. The connector is received by the connector receiver in the charge port to align the opening with the charge port receptacle. A frame extends about the peripheral edge of the base member. The frame includes a plurality of walls that project outwardly of the first surface of the base member. The plurality of walls and the first surface of the base member forming a recess.

(12) In addition to one or more of the features described herein the plurality of walls include a first wall, a second wall, a first side wall connecting the first wall and the second wall and a second side wall connecting the first wall and the second wall, the second wall, first side wall, and the second side wall extending from the first surface a first distance and the first wall extending from the first surface a second distance that is greater than the first distance.

(13) In addition to one or more of the features described herein the first wall defines an upper wall and the second wall defines a lower wall.

(14) In addition to one or more of the features described herein an external charging device is electrically connected to the charge port receptacle at an interface, the upper wall extending from the base member over the interface.

(15) In addition to one or more of the features described herein the external charging device includes a handle, the upper wall extending over a portion of the handle.

(16) In addition to one or more of the features described herein each of the first wall, the second wall, the first side wall, the second side wall, and the first surface of the base member form a cavity receptive of the external charging device.

(17) In addition to one or more of the features described herein the connector receiver comprises a quick charge port.

(18) In addition to one or more of the features described herein the connector comprises a dummy quick charge connector supported on the external charging device.

(19) In addition to one or more of the features described herein the external charging device includes a charge plug connected to the charge port receptacle.

(20) In addition to one or more of the features described herein the charge plug extends through the opening into the charge port receptacle.

(21) In addition to one or more of the features described herein each of the first wall, the second wall, the first side wall, and the second side wall extend from the first surface at an angle.

(22) In addition to one or more of the features described herein each of the first wall, the second wall, the first side wall, and the second side wall extend from the base member at an obtuse angle relative to the first surface.

(23) The above features and advantages, and other features and advantages of the disclosure are readily apparent from the following detailed description when taken in connection with the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Other features, advantages and details appear, by way of example only, in the following detailed description, the detailed description referring to the drawings in which:

(2) FIG. 1 is a left side view of a vehicle including a charge port weather shield, in accordance with a non-limiting example;

(3) FIG. 2 is a disassembled view of the charge port weather shield prior to mounting to a charge port of the vehicle of FIG. 1, in accordance with a non-limiting example;

(4) FIG. 3 is a side view of the charge port weather shield and charge connector, in accordance with a non-limiting example;

(5) FIG. 4 is a front plan view of the charge port weather shield of FIG. 3, in accordance with a non-limiting example; and

(6) FIG. 5 is a rear plan view of the charge port weather shield of FIG. 3, in accordance with a non-limiting example.

DETAILED DESCRIPTION

(7) The following description is merely exemplary in nature and is not intended to limit the present disclosure, its application or uses. It should be understood that throughout the drawings, corresponding reference numerals indicate like or corresponding parts and features.

(8) A vehicle, in accordance with a non-limiting example, is indicated generally at **10** in FIG. 1. Vehicle **10** includes a body **12** supported on a frame (not shown) and a plurality of wheels **16**. Body **12** includes a passenger compartment **20** that includes a driver's seat **22** and at least one rear passenger seat **24**. Vehicle **10** takes the form of an electric vehicle (EV) **28** including a charge port **30** having a door **32** that is selectively pivotally mounted to body **12**. Charge port **30** includes a charge port receptacle **33** and a fast charge port **34**, FIG. 2, that may form a connector, receiver **35** as will be discussed herein. Charge port receptacle **33** and fast charge port **34** are connected to a rechargeable energy storage system (RESS) **36** supported in body **12**. RESS **36** is operatively connected to an electric motor **38** that provides motive power to vehicle **10**.

(9) As shown in FIG. 2, an external charging device **40** is selectively connected to charge port **30**. External charging device **40** forms part of a charging station (not shown) that, when connected to charge port **30**, provides electric energy to RESS **36**. External charging device **40** includes a handle **42** that supports a charging connector **44**. A charge port shield **58** may serve as an interface between external charging device **40** and charge port **30**. As will be detailed more fully herein, charge port shield **58** protects charge port **30** from external elements such as rain, snow, and the like.

(10) Referring to FIG. 3 and with continued reference to FIG. 2, charge port shield **58** includes a base member **60** and a frame **62**. Base member **60** includes a peripheral edge **66** defining a first surface **68** and a second surface **70** that is opposite first surface **68**. An opening **72** passes through base member **60**. As will be detailed more fully herein, opening **72** is receptive of charging connector **44**. In a non-limiting example, a connector **76** projects outwardly from second surface **70**. Connector **76** takes the form of a dummy fast charging connector **78**. Dummy fast charging connector **78** does not include any electrical connections and may be received by connector receiver **35** of fast charge port **34**. Connector **76** selectively joins charge port shield **58** to charge port **30**.

(11) In a non-limiting example, frame **62** is formed by a plurality of walls **90** that extend about

peripheral edge **66** of base member **60**. Plurality of walls **90** include a first wall **92**, a second wall **94**, a first side wall **96**, and a second side wall **98**. First side wall **96** and second side wall **98** extend between and connect with first wall **92** and second wall **94**. In a non-limiting example, when charge port shield **58** is connected to charge port **30**, first wall **92** is a top or upper wall **104** and second wall **94** is a bottom or lower wall **106**.

(12) In a non-limiting example, each of the plurality of walls **90** extend at an angle from base member **60** so as to form a recess or cup-like structure. In a non-limiting example, each of the plurality of walls **90** extend at an obtuse angle relative to first surface **68** from peripheral edge **66** of base member **60**. In a non-limiting example, each of the second wall **94**, the first side wall **96** and the second side wall **98** extend from base member **60** a first distance, and first wall **92** extends from base member **60** a second distance that is greater than the first distance. In this manner, first wall **92** forms a cover or hood **114** that protects charge port receptacle **33** from weather and other contaminants when charging outdoors.

(13) In a non-limiting example, charge port shield **58** is secured to vehicle **10** by inserting connector **76** into connector receiver **35** of charge port **30**. When connected, opening **72** is aligned with and exposes charge port receptacle **33**. At this point, external charging device **40** may be connected to charge port **30** by inserting charging connector **44** through opening **72** into charge port receptacle **33** forming an electrical interface. In a non-limiting example, first wall **92** extends over the electrical interface. In another non-limiting example, first wall **92** extends from base member **60** over a portion of handle **42** in order to protect the electrical interface from external elements such as rain, snow, dirt and the like.

(14) The terms “a” and “an” do not denote a limitation of quantity, but rather denote the presence of at least one of the referenced item. The term “or” means “and/or” unless clearly indicated otherwise by context. Reference throughout the specification to “an aspect”, means that a particular element (e.g., feature, structure, step, or characteristic) described in connection with the aspect is included in at least one aspect described herein, and may or may not be present in other aspects. In addition, it is to be understood that the described elements may be combined in any suitable manner in the various aspects.

(15) When an element such as a layer, film, region, or substrate is referred to as being “on” another element, it can be directly on the other element or intervening elements may also be present. In contrast, when an element is referred to as being “directly on” another element, there are no intervening elements present.

(16) Unless specified to the contrary herein, all test standards are the most recent standard in effect as of the filing date of this application, or, if priority is claimed, the filing date of the earliest priority application in which the test standard appears.

(17) Unless defined otherwise, technical and scientific terms used herein have the same meaning as is commonly understood by one of skill in the art to which this disclosure belongs.

(18) While the above disclosure has been described with reference to exemplary embodiments, it will be understood by those skilled in the art that various changes may be made and equivalents may be substituted for elements thereof without departing from its scope. In addition, many modifications may be made to adapt a particular situation or material to the teachings of the disclosure without departing from the essential scope thereof. Therefore, it is intended that the present disclosure not be limited to the particular embodiments disclosed, but will include all embodiments falling within the scope thereof.

Claims

1. A vehicle charge system comprising: a charge port including a charge port receptacle and a connector receiver, the charge port receptacle being configured to transfer electrical energy from an electrical source to a rechargeable electric storage system (RESS); and a charge port shield

supported at the charge port, the charge port shield comprising: a base member having a peripheral edge defining a first surface and a second surface opposite the first surface, the base member including an opening alignable with the charge port receptacle and a connector extending outwardly of the second surface, the connector being received by the connector receiver in the charge port to align the opening with the charge port receptacle; and a frame extending about the peripheral edge of the base member, the frame including a plurality of walls that project outwardly of the first surface of the base member, the plurality of walls and the first surface of the base member forming a recess, wherein the connector receiver comprises a quick charge port, and wherein the connector comprises a dummy quick charge connector.

2. A vehicle charge system comprising: a charge port including a charge port receptacle and a connector receiver, the charge port receptacle being configured to transfer electrical energy from an electrical source to a rechargeable electric storage system (RESS); and a charge port shield supported at the charge port, the charge port shield comprising: a base member having a peripheral edge defining a first surface and a second surface opposite the first surface, the base member including an opening alignable with the charge port receptacle and a connector extending outwardly of the second surface, the connector being received by the connector receiver in the charge port to align the opening with the charge port receptacle; and a frame extending about the peripheral edge of the base member, the frame including a plurality of walls that project outwardly of the first surface of the base member, the plurality of walls and the first surface of the base member forming a recess, wherein the plurality of walls include a first wall, a second wall, a first side wall connecting the first wall and the second wall and a second side wall connecting the first wall and the second wall, the second wall, the first side wall, and the second side wall extending from the first surface a first distance and the first wall extending from the first surface a second distance that is greater than the first distance.

3. The vehicle charge system according to claim 2, wherein the first wall defines an upper wall and the second wall defines a lower wall.

4. The vehicle charge system according to claim 2, wherein each of the first wall, the second wall, the first side wall, and the second side wall extend from the first surface at an angle.

5. The vehicle charge system according to claim 4, wherein each of the first wall, the second wall, the first side wall, and the second side wall extend from the base member at an obtuse angle relative to the first surface.

6. The vehicle charge system according to claim 4, wherein each of the first wall, the second wall, the first side wall, the second side wall, and the first surface of the base member form an external charging device receiving cavity.

7. The vehicle charge system according to claim 2, wherein the connector receiver comprises a quick charge port.

8. The vehicle charge system according to claim 7, wherein the connector comprises a dummy quick charge connector.

9. A vehicle comprising: a body defining a passenger compartment; an electric motor supported in the body; a rechargeable energy storage system (RESS) supported at the body; a charge port supported at the body, the charge port including a charge port receptacle connected to the RESS and a connector receiver; and a charge port shield supported at the charge port, the charge port shield comprising: a base member having a peripheral edge defining a first surface and a second surface opposite the first surface, the base member including an opening alignable with the charge port receptacle and a connector extending outwardly of the second surface, the connector being received by the connector receiver in the charge port to align the opening with the charge port receptacle; and a frame extending about the peripheral edge of the base member, the frame including a plurality of walls that project outwardly of the first surface of the base member, the plurality of walls and the first surface of the base member forming a recess, wherein the plurality of walls include a first wall, a second wall, a first side wall connecting the first wall and the second

wall and a second side wall connecting the first wall and the second wall, the second wall, first side wall, and the second side wall extending from the first surface a first distance and the first wall extending from the first surface a second distance that is greater than the first distance.

10. The vehicle according to claim 9, wherein the first wall defines an upper wall and the second wall defines a lower wall.

11. The vehicle according to claim 10, further comprising an external charging device electrically connected to the charge port receptacle at an interface, the upper wall extending from the base member over the interface.

12. The vehicle according to claim 11, wherein the external charging device includes a handle, the upper wall extending over a portion of the handle.

13. The vehicle according to claim 11, wherein each of the first wall, the second wall, the first side wall, the second side wall, and the first surface of the base member form a cavity receptive of the external charging device.

14. The vehicle according to claim 11, wherein the connector receiver comprises a quick charge port.

15. The vehicle according to claim 14, wherein the connector comprises a dummy quick charge connector supported on the external charging device.

16. The vehicle according to claim 11, wherein the external charging device includes a charge plug connected to the charge port receptacle.

17. The vehicle according to claim 16, wherein the charge plug extends through the opening into the charge port receptacle.

18. The vehicle according to claim 9, wherein each of the first wall, the second wall, the first side wall, and the second side wall extend from the first surface at an angle.

19. The vehicle according to claim 9, wherein each of the first wall, the second wall, the first side wall, and the second side wall extend from the base member at an obtuse angle relative to the first surface.

20. The vehicle according to claim 9, wherein the connector receiver comprises a quick charge port, and wherein the connector comprises a dummy quick charge connector.
